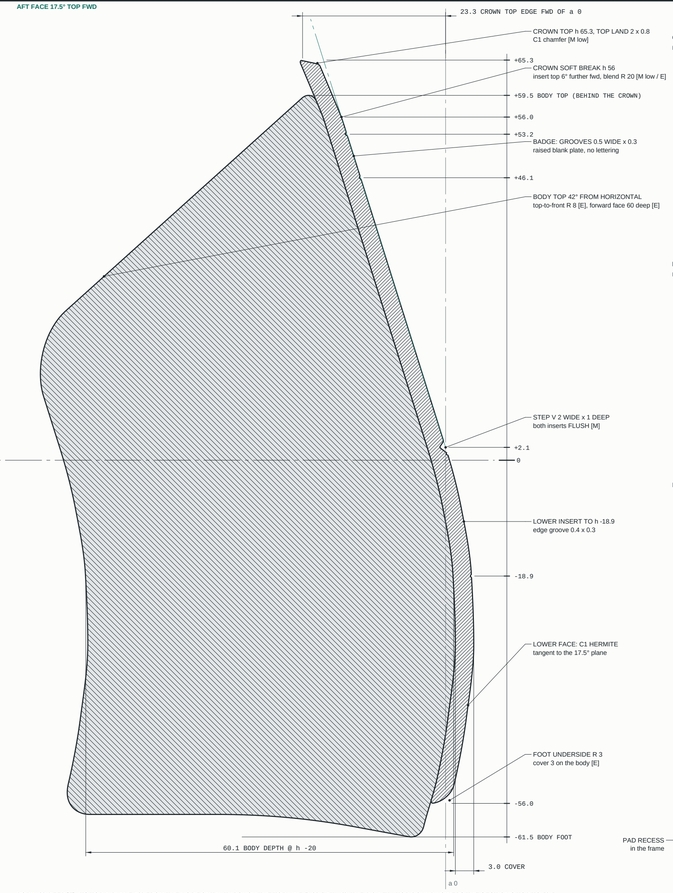
AFT FACE 17.5° TOP FWD

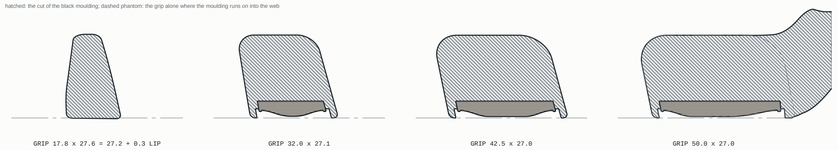


BODY: 60 DEEP [E], ITS TOP A PLANE 42° FROM THE HORIZONTAL THROUGH THE TOP LINE (h 60.5 AT u 0); THE R 2 AFT x TOP ROUND LOWERS THE SECTION'S TOP TO 59.5 rounds in order: aft R 2; top R 2; forward R 8, foot R 5 [E]; one moulding with the grips (R 3)
PLANE u 0 (CUTTING PLANE Z-Z ON L6Y FRONT ELEVATION)

SECTION Z-Z

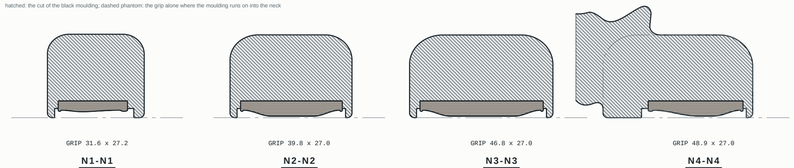
YOKE CENTRELINE u 0, FROM OUTBOARD, NOSE LEFT - 2:1

OUTBOARD GRIP, NORMAL TO THE AXIS, LOOKING DOWN IT (FWD UP, PAD DOWN; OUTBOARD LEFT)
hatched: the cut of the black moulding; dashed phantom: the grip alone where the moulding runs on into the web



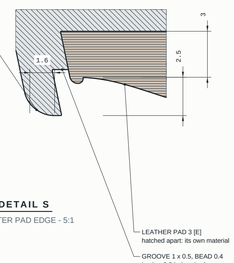
GRIP 17.8 x 27.6 = 27.2 + 0.3 LIP — **A1-A1** — h -30 - 1:1
GRIP 32.0 x 27.1 — **A2-A2** — h -10 - 1:1
GRIP 42.5 x 27.0 — **A3-A3** — h 14 - 1:1
GRIP 50.0 x 27.0 — **A4-A4** — h 40 - 1:1

INBOARD GRIP, NORMAL TO THE AXIS, LOOKING DOWN IT (FWD UP, PAD DOWN; INBOARD RIGHT)
hatched: the cut of the black moulding; dashed phantom: the grip alone where the moulding runs on into the neck

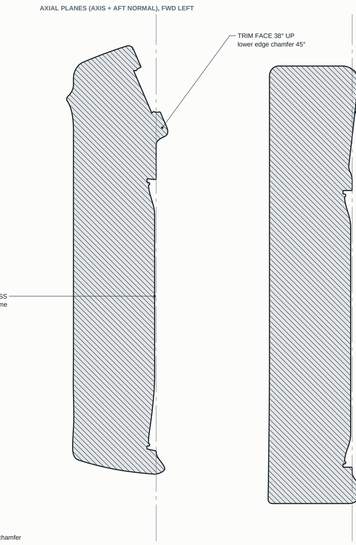


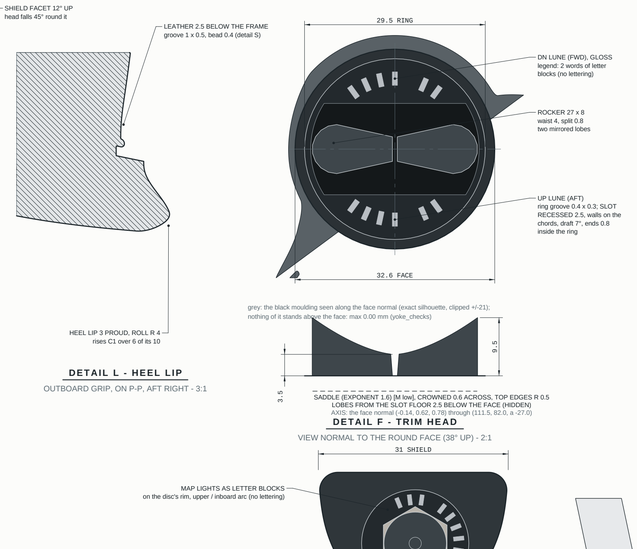
GRIP 31.6 x 27.2 — **N1-N1** — h -30 - 1:1
GRIP 39.8 x 27.0 — **N2-N2** — h -10 - 1:1
GRIP 46.8 x 27.0 — **N3-N3** — h 14 - 1:1
GRIP 48.9 x 27.0 — **N4-N4** — h 40 - 1:1

PAD: leather 2.5 below the 2 frame, crowned to 0.5 below over 8; groove 1 x 0.5 + bead 0.4; frame roll R 1.5; forward R 11 [E] <= 0.45 x half-width; depth 27 (grip_depth) [M]



DETAIL S
A3 OUTER PAD EDGE - 5:1



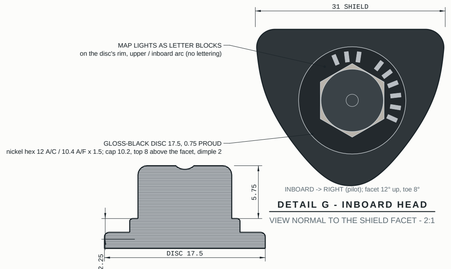


DETAIL L - HEEL LIP
OUTBOARD GRIP, ON P-P, AFT RIGHT - 3:1

grey: the black moulding seen along the face normal (exact silhouette, clipped +/-21);
nothing of it stands above the face: max 0.00 mm (yoke_checks)

SADDLE (EXPONENT 1.6) [M low], CROWNED 0.6 ACROSS, TOP EDGES R 0.5
LOBES FROM THE SLOT FLOOR 2.5 BELOW THE FACE (HIDDEN)
AXIS: the face normal (-0.14, 0.62, 0.78) through (111.5, 82.0, a -27.0)

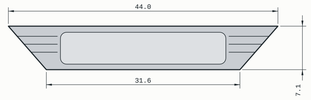
DETAIL F - TRIM HEAD
VIEW NORMAL TO THE ROUND FACE (38° UP) - 2:1



INBOARD -> RIGHT (pilot); facet 12° up, toe 8°

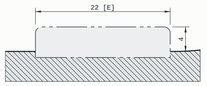
DETAIL G - INBOARD HEAD
VIEW NORMAL TO THE SHIELD FACET - 2:1

DISC 0.75 PROUD R 0.5; HEX 10.4 A/F x 1.5; CAP 10.2, TOP R 1, DIMPLE 2 x 0.4 [E]
BUTTON: EXACT CUT ALONG THE FACET NORMAL - 3:1
AXIS: the facet normal through (-106.3, 76.0, a -23.9); facet point (-106, 75.6, a -23.8)



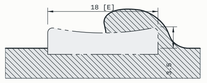
RAISED BLANK PLATE 27.0 x 5.2, 0.2 PROUD, R 1 / EDGE R 0.2 (NO LETTERING); GROOVES 0.5 x 0.3 CUT IN THE INSERT

DETAIL H - BADGE
ABSTRACT WINGED PLATE, TOP AT h 53.2 - 2:1



h 56 ON THE FORWARD FACE, 10 WIDE, TOP R 1, FLAT TOP
AXIS: THE FORWARD FACE'S NORMAL THROUGH (106.9, 52.2, -43.2)

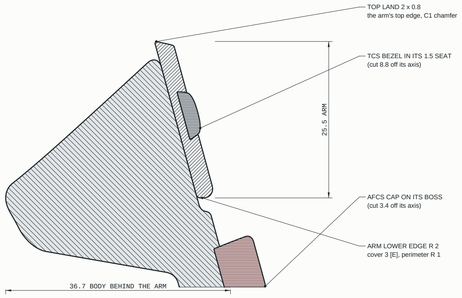
DETAIL W1 - OUTBOARD: PTT TRIGGER [E]
PLACEHOLDER (PHANTOM), NEVER SEEN; FWD UP, GRIP AXIS ACROSS - 2:1



h 50 ON THE FORWARD FACE, 10 WIDE, TOP R 1, SADDLE 1
AXIS: THE FORWARD FACE'S NORMAL THROUGH (-99.4, 41.2, -39.0)

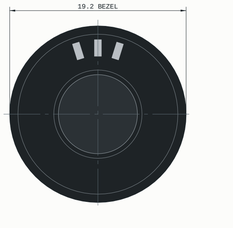
DETAIL W2 - INBOARD: CHECKLIST ROCKER [E]
PLACEHOLDER (PHANTOM), NEVER SEEN; FWD UP, GRIP AXIS ACROSS - 2:1

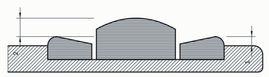
UP-INBOARD



SECTION K-K

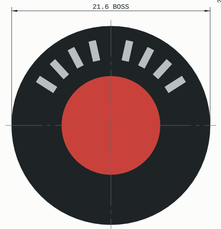
THROUGH (52, 45) NORMAL TO THE ARM (30° UP OUTBOARD), LOOKING INBOARD (L6Y FRONT ELEVATION) - 2:1

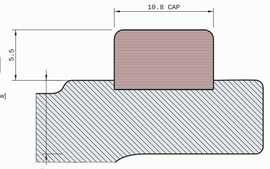




bezel 1 proud, dish r 4.8-8.7 0.8 deep [E], button dome 1.2, its top 2 ABOVE the bezel [M low]
AXIS: the 17.5° face's normal through (58.5, 51.4, a -15.5)

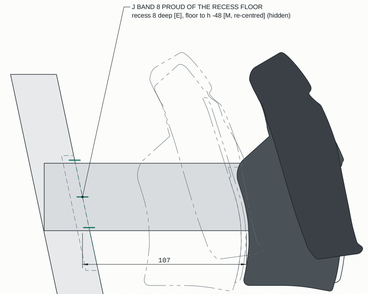
DETAIL D - TCS
u 58.5 h 51.4 ON THE WHITE ARM; PROFILE = EXACT CUT - 3:1





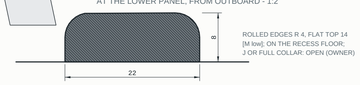
boss 1.5 proud of the body [E], 8 deep: its back free over the hand slot; edge R 0.8, cap top R 1
AXIS: the 17.5° face's normal through the boss face (60.6, 23.4, a -8.2) = face - cover 3 + 1.5

DETAIL E - AFCS DISCONNECT
u 60.6 h 23.4; BOSS = A SOLID OF THE MOULDING; PROFILE = EXACT CUT - 3:1



PHANTOM: FULL PUSH -65 (OWNER 2026-10-11; WAS -90 [E]), leaves 42 to the panel

DETAIL J - COLUMN, SIDE
AT THE LOWER PANEL, FROM OUTBOARD - 1:2

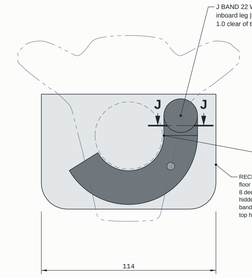


ROLLED EDGES R 4, FLAT TOP 14
[M low]; ON THE RECESS FLOOR;
J OR FULL COLLAR: OPEN (OWNER)

SECTION J-J - BAND

ACROSS THE VERTICAL LEG (P1046410 / 09) - 2:1

OWNER DECISION 2026-10-11 (fit_check 30c):
THE J BAND AND THE U RECESS RE-CENTRED ON THE COLUMN AXIS = THE HUB (WL 1845; ROLL PIVOT AND KNEE RESULTS KEPT). REASON: THE PHOTOS READ THE JOINT LINE, I.E. THE COLUMN AXIS, AT WL 1821 +/-10, 24 UNDER THE HUB: BAND AND RECESS DRAWN 24 UP WITH THE AXIS



phantom: the shell; hidden: the column dia 44 (satin black)

DETAIL J - SURROUND, FRONT
LOOKING FORWARD (PILOT, INBOARD RIGHT) - 1:2

PHOTO CHECK (rev M): THE DRAWN SOLIDS RAY-CAST THROUGH THE PHOTO CAMERAS (mm, + = drawn wider)

PHOTO (CAMERA)	COVER LM	HEADS	GRIP OUT: OUTER / SLOT (mean, max)	GRIP IN: OUTER / SLOT (mean, max)	PADS mean (max)
P1046408 pilot (panel_408)	1.4 / 2.7	2.1 / 1.4	+0.2, +3.9 / +1.0, +5.0	+0.4, -5.8 / -1.8, -4.2	out 0.7 (2.5); in 0.6 (1.3)
P1046408 co-pilot (panel_408)	1.8 / 3.9	2.9 / 1.0	+1.1, +6.6 / +0.7, +2.1	-0.4, -2.8 / +0.1, +3.3	out 1.7 (3.4); in 0.8 (1.9)
P1046410 pilot	0.8 / 1.6	5.4 / 0.7	-0.5, -6.9 / +0.2, -6.1	-0.6, -7.6 / +2.9, +6.2	-
Mexico press, pilot	1.7 / 3.9	1.7 / 2.3	+2.3, +5.5 / +0.7, +1.2	-1.9, -7.8 / +0.4, +1.3	in 1.9 (5.1)
Mexico press, co-pilot	1.9 / 3.1	0.9 / 0.6	+0.3, +6.7 / +0.5, +2.6	+1.7, +4.8 / -3.9, -6.8	out 1.1 (2.7)
P1046409 co-pilot (two-view camera)	1.0 / 2.1	6.8 / 2.7	-2.1, -7.7 / +0.5, +9.8	+0.7, +7.8 / -2.9, -7.4	out 1.7 (4.3); in 3.4 (6.9)
rev L, P1046408 pilot (judges r4)	1.4 / 2.7	2.1 / 1.4	+0.3, +4.6 / +1.1, +6.6	+0.4, +6.2 / -1.8, -4.2	out 0.7; in 0.6

GRIP SIDES PER BAND OF h: SIGNED MAX mm (h -50..-30 | -30..-10 | -10..10 | 10..30)

SIDE	P408 L	P408 R	P410 L	MEX L	MEX R	P409 R
OUT SLOT	+5.0 +1.9 +1.1 +0.4	+1.9 +2.1 +0.8 +0.5	+5.2 +1.9 -1.3 -6.1	. . +1.2 +0.9	+2.6 +2.3 -1.0 -1.6	+9.8 +1.6 -0.8 -5.1
OUT OUTER	+2.8 -0.8 -0.3 -0.8	-0.4 +3.1 -1.5 -0.6	-4.4 +3.1 -1.5 -0.6	. . . .	-2.1 . -5.5 -0.7	-7.1 +5.8 +4.7 -7.7
IN SLOT	-2.5 -4.2 -4.1 -1.8	-0.8 -2.3 +1.8 +3.3	+2.4 +3.4 +5.4 +6.2	. +0.3 +1.3 +1.0	. -5.3 -3.2 -6.8	-7.4 -5.0 -7.3 +1.1
IN OUTER	-5.8 -5.2 -1.1 +0.9	-2.8 -2.1 -0.9 -1.0	-5.3 -7.6 -6.5 +1.2	+6.4 -7.8 -6.5 -7.2	+4.8 +2.0 +2.5 +2.2	+7.8 -6.2 -7.0 -6.5

GRIP PLANES (rev L, judges FID3-01): rake / toe / aD / walls / depth fitted per grip to the curated rows and pad traces of 6 views – P1046408 L / R, P1046410, Mexico L / R and P1046409 (posed this round: a two-view camera with panel_408, 25° from inboard) -- with the front outline held: outboard 20.4° / 6.9° / -4.4 / 0.64, depth 30 / 27 / 22 at h +47 / 0 / -45; inboard 21.3° / 18.7° / -6.1 / 0.04 (its depth above h 0 unconstrained: no view sees the inboard forward face). Table: 20 / 18 / -3.75, -6.0 / 0.6, 0 / 27.5 at the bottom, 27 above. Outboard cost +7 % at rake -2.5°, +8 % at aD +/-5, +10 % at a bottom depth +/-4; the TOE is weak (7..22 within +7 %, inboard 0..26 within the noise; open point 2 of L6Y). The AOPA views are CHECK ONLY (one radial term; the co-pilot outer -11 / -22, slot -12 / -15, pad 9.5): not fitted, not tabled. Rev M (judges FID4-05): that co-pilot camera is EXTRAPOLATED from the pilot-yoke solution -- not a degenerate pose; its re-calibration on the panel round the co-pilot yoke is open; the toe / depth law / head depths stay [M weak].

INBOARD NECK (criterion, P1046408): the drawn moulding's top edge vs the strongest photo edge per pixel column, arm tip to head -- pilot mean -1.3, |mean| 2.3, max -7.4 mm (against the red placard; the max where the edge turns vertical under the head, ill-posed per column); co-pilot -4.6, max -7.5: there the search locks onto the PFD's text rows above the neck (overlay inspected: the drawn edge lies on the photo's neck, dark on dark). Rev L raised the last three samples 0.6-2.2 mm (body_in).

Pose per photo fitted to the COVER landmarks only (TCS, AFCS, badge, step ends, lower-insert corners, foot; P1046409 with its own picks): heads, grips, pads, neck and shell are checks. Grip rows: the CURATED photo silhouettes -- auto edges per image row in the near views with the panel recesses masked, the inboard slot sides hand-traced and snapped past the J band / panel seam, P1046409 traced at 4x; slot rows above h 24 (the arm's underside, not the grip) not scored -- against the ray-cast grip's own hits (part-labelled), per image row, h on the grip axis. The band maxima below h -27 are the rows that cut the grips' rounded bottom ends (1 mm of height reads 3-10 mm of width there); above it the row medians are 0.5-0.7 (outboard slot), 0.3-2.6 (outer sides), 0.4-3.5 (inboard slot), 90 % of the rows within 1-6.6. Overlays: refs/cache/overlays/yoke (private).

CORNERS & RADII (every other sample is on a C2 spline); NO OTHER EDGE IS SHARP

OUTLINE	N	CORNERS: SAMPLE #, (u, h), RADIUS (curve_law)
shell (white cover), half	49	#4 (22, 62.8) R 3; #10 (33.7, 52) R 1.5; #21 (66, 60.2) R 2; #22 (72.3, 54.8) R 2; #23 (72.3, 47.8) R 2; #24 (69, 44.6) R 2; #45 (20, -55.3) R 6
lower insert, half	8	#0 (0, 1) R 0; #1 (12.6, 3) R 0.5
outboard grip aft face	38	#6 (130.5, 93.6) R 4; #20 (136.6, -30.4) R 5; #23 (121, -35.5) R 9
inboard grip aft face	37	#2 (91.4, 84) R 4; #6 (118.5, 87.6) R 6; #10 (125.6, 60) R 3; #21 (140.5, -36) R 5; #24 (118.5, -45.8) R 4
outboard pad	29	#0 (125.5, 60) R 3; #7 (91, 38) R 5; #16 (123.5, -28.4) R 3; #19 (134.8, -25.5) R 4
inboard pad	31	#0 (127.8, 55.3) R 3; #7 (89.4, 32) R 4; #17 (116, -34.4) R 3; #21 (139, -29.6) R 4
shield facet	9	#0 (-15.5, 12) R 3; #1 (15.5, 12) R 3
black body, outboard half	37	NO CORNERS: one C2 run joined to the other half at u 0 (rev M, judges DR4-07); its first 12 samples are the top edge (body_top)
black body, inboard half	35	NO CORNERS: one C2 run joined to the other half at u 0 (rev M, judges DR4-07); its first 11 samples are the top edge (body_top)
insert corners	-	junction R 0; step_end R 0.5
cover edges	-	perimeter R 1; arm lower R 2; foot underside R 3; tip facet corners R 2
grip edges	-	frame roll R 1.5; forward R 11 [E] capped at 0.45 x the inscribed half-width; heel R 4; trim face R 1.5; pad step crisp, groove 1 x 0.5, bead 0.4
body edges	-	rounds in order: sides x aft face R 2, x top plane R 2, x forward face R 8 [E] easing to R 5 under the foot, R 3 behind the arms / necks [E]; under the foot the side draft's offset capped at 4 [E]; AFCS boss aft edge R 0.8, back edge R 0.8, blend R 1.5
moulding	-	body + grips joined by a ROLLING-BALL round R 3 (none where their faces run flush); then each pad's pocket, groove and bead (pad_finish); the web's aft face on the trim face's plane 3.5 below it, joined to the grip plane over 8; the low neck's aft face onto the shield facet's plane 0.5 below it over |u| 76-90; body top line a C2 cubic
head zone out / in	-	aft side: prism trimmed 3 inside the head above h 67 / 61, drafted 3 / 1 per mm below from a bend of R 4, cut R 1.5; head chamfer R 3, floor R 3 at h 62 / 50; head-to-prism R 3; then walls (aft edge R 1.5 -> 2) and forward face (R 11 capped -> 3; over the head's top onto it R 3); shield facet edge R 1.5, its directrix tangent at the widest points (no corner)
cover features	-	crown soft break R 20; top land C1 chamfer 2 x 0.8; TCS seat 1.5 deep

NOTES

1. CURVE LAW (rev M text, judges DR4-03): each outline split at its corner samples into runs, each run a C2 cubic spline on centripetal knots ('natural' ends at corners, periodic for a closed outline without corners), 0.05 sampling (arcs too); corners R 0 sharp, R > 0 a tangent fillet (cubic Bezier arc; tangent length R tan(θ/2) capped at 0.45 of the shorter run -- none capped); 'YOKE_DESIGN curve_law; the solids use the EXACT distance to these polylines (beyond 0.3 mm a C2 B-spline of exact node distances, blended C2).
2. ACCEPTANCE (model/yoke_checks.py, judges DR2-01..04, DR4-01 / 02): fields: zero level vs the curve-law polylines, max shell 0.0059, grip_out 0.0045, grip_in 0.0079, pad_out 0.0045, pad_in 0.0045, body 0.0052 mm; curve law (C2): max relative curvature jump at the interior samples 8.3e-07 over 301 samples of 10 outlines; nothing above the trim face: max 0.00 mm; above the shield facet: max 0.00 mm; untabled cusps (contour turning > 25 deg within +-0.15 mm AND the surface normals turning as much): grip_out 0 / 54 sections, grip_in 0 / 54 sections, necks / web / heads (u planes) 0 / 19 sections, horizontal (h planes) 0 / 16 sections, Z-Z 0, K-K 0 (52 contour tips set aside: planes grazing a tabled round, the normals turning less); black moulding one piece per side in every horizontal section h 62..94 (step 2; none above a head's top): yes; CREASES (rev M, judges DR4-02; adaptive scan of 14 views incl. sides / below: every pixel pair turning > 4 deg refined 2 x 16x, a crease when the turn stays in one sub-step): untabled 0; 2218 tabled crease pixels set aside, 86562 candidate pairs resolved as rounds; joins read smooth fields at the rounds' own surface points (a ridge = a unit gradient turning > 20 deg within 0.01 mm; min |grad| in brackets): body x outboard grip (R 3) 6 / 6934 points ([grad| >= 0.59), body x inboard grip (R 3) 0 / 3023 points ([grad| >= 0.70), outboard prism aft side x head (R 3) 1 / 2593 points ([grad| >= 0.56), inboard prism aft side x head (R 3) 0 / 1510 points ([grad| >= 0.94); pads EXPOSED as drawn (judges DR4-01; the moulding ray-cast from aft vs the drawn leather / groove / head heights inside each pad outline): out 0.0 mm2 off by > 0.05 (max 0.00), buried 0.0 mm2 off by > 0.05 (max 0.00), buried 0.0 mm2; fillets reduced by the 0.45-of-run cap (judges DR4-03): none; vertex normals vs analytic (depth-map meshes 0.25 mm from aft, 0.4 from above-aft / the front; 95th / 99th / max deg; tabled creases, the cover's 0.8 mm top land and part boundaries apart): cover stem / inserts 0.00 / 0.06 / 0.7 (creases 21); cover arm + TCS 0.15 / 0.79 / 4.3 (creases 8); outboard grip pad / frame 0.36 / 1.19 / 29.4 (creases 2); inboard grip pad / frame 0.31 / 1.54 / 4.2 (creases 30); outboard head + face 0.73 / 1.92 / 5.4 (creases 73); inboard head + shield 0.65 / 1.57 / 5.3 (creases 4); web (aft) 0.46 / 1.83 / 30.3 (creases 6); body top + web top (from above-aft) 0.01 / 0.85 / 12.0 (creases 6); forward faces (from the front) 0.43 / 1.35 / 10.1 (creases 10).
3. SECTIONS: exact contours of the solids (model/yoke_geom.py) on the cutting planes; the black body and the grips are one moulding (a rolling-ball round R 3), the white cover a separate part, the leather pad hatched apart (its 3.0 layer [E]); where a section is clipped to its window the cut edge is a thin phantom break line.
4. Sources per value: the [M] / [E] / [S] tags of YOKE_DESIGN (printed in model/interior.py).

REVISIONS - PC12-L6Y2

REV	DATE	DESCRIPTION
A	2026-10-05	first issue with L6Y rev B (judges r1): exact sections of the rev J solids, grip profiles with the heel lip, trim face / shield, switch profiles, column surround, corner table, photo check
B	2026-10-05	rev K (judges r2): exact-field solids; Z-Z / K-K with ordinates; cusp-free grip sections; pad groove + bead, heel lip R 4; D / E / G exact switch cuts; J-J; acceptance; photo check on curated silhouettes, grip planes fitted to 7 views
C	2026-10-05	rev L (judges r3): K-K through TCS / AFCS; grip sections cut the moulding; dimensions off the solids; head-zone surfaces and rounds tabled; acceptance on heads / necks / web / tops / forward faces + one-piece and C2 rows; photo check on 6 views; detail J U recess
D	2026-10-05	rev M (judges r4): pads exposed as drawn (pocket cut after the join); rolling-ball R 3 joins; head zones share walls / forward face; leather hatched apart; detail S, break lines, detail W [E] switches; adaptive crease / ridge / pad acceptance, all clean (head top R 3 onto the forward face, body sides x top rolling ball, C2 forward-radius cap, tangent inboard direction, 0.05 sampling, C2 fields); photo check re-run
E	2026-10-11	rev N (3-D judges r1): F rocker slot 2.5 deep (draft 7 deg), letter-block legends in the pilot's reading order; G MAP LIGHTS blocks, satin-nickel collar; H raised blank plate; J recess to h 27, band 22 wide; OWNER DECISIONS 2026-10-11: J band + recess re-centred on the column axis (hub WL 1845; photos 1821), push 65 (was 90 [E])
F	2026-10-11	rev O (3-D judges r2): G push button = gloss-black legend disc 17.5 + nickel hex 12 A/C + cap 10.2, MAP LIGHTS on the disc; F rocker ends 9.5; crowned 0.6; H one outline groove, plate 0.2; legends 0.35 of the pitch

PILATUS PC-12 PRO

PC-24-STYLE CONTROL YOKE - SECTIONS & DETAILS

FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L6Y2	F	AS SHOWN	A1 · 2 OF 2	mm

DRAWN	DATE
Claude - loftkit parametric model	2026-10-11

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION